

ToAST: Toy Setting for the Anchor Token Hypothesis

Liyang Deng, Elizaveta Dovedova, Magdalena Ljubić, Yuwen Peng

Department of Computational Linguistics, Heidelberg University

<https://github.com/liyangdeng/toy-setting-for-anchor-tokens>

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Abstract

Cross-lingual transfer in language models has been credited to several shared properties that are entangled in natural data. We disentangle them with pairs of artificial languages built from a shared knowledge graph, generated by a PCFG with switch-controlled word order. This setup allows us to vary lexical overlap, structure, shared semantics, graph density, and sentence-boundary tokens independently. We train small BERT models and measure alignment with four metrics: word translation, sentence retrieval, LM-head prediction, and linear probing. We find that shared tokens are not required for transfer: with no lexical overlap, transfer emerges when the two languages have identical structure, but not at lower levels of structural similarity. Lexical overlap, semantic overlap, and graph density generally improve alignment. Finally, linear probing detects cross-lingual information in the model’s hidden representations that is not reflected in LM-head predictions.

1 Introduction

Modern language models are routinely pre-trained on text from many languages at once. Despite receiving no language labels or parallel data, models such as mBERT (Devlin et al., 2019) can learn a task from labelled data in one language and apply it to another without labelled examples in the target language. This ability is known as *cross-lingual transfer* (Pires et al., 2019; Wu and Dredze, 2019). Why this transfer emerges, and which shared properties are responsible, has been debated (Section 2). In natural languages, properties such as lexical overlap, script, and word order are often entangled, making their individual contributions difficult to isolate.

We therefore construct pairs of artificial languages from a shared semantic backbone, with independent control over these properties.

We take the so-called *anchor token hypothesis* (Pires et al., 2019; Conneau et al., 2020) as a starting point by first considering whether shared tokens between the two languages help realize cross-lingual transfer, and then extend to broader properties, including structural, semantic overlap and graph density, which captures how richly the concepts are connected through different relationships, the last two of which to our knowledge have not been examined before. We measure the resulting cross-lingual alignment with several complementary measures, defined in Section 4.

2 Related Work

Prior work has debated what drives cross-lingual transfer. Pires et al. (2019) find that mBERT transfers across scripts, more so between languages of similar word order, while Karthikeyan et al. (2020), Conneau et al. (2020), and Artetxe et al. (2020) show that it persists with little or no shared vocabulary, pointing to structure over the lexicon. Two studies shape our method: Dufter and Schütze (2020) study BERT’s multilinguality in a small controlled setup; White and Cotterell (2021) isolate typological factors with artificial languages and grammar-level switches.

3 Artificial Language Generation

3.1 Knowledge Graph

To study cross-lingual alignment in a controlled way, we need a semantic world where concepts and relations can be explicitly controlled. Therefore, we constructed a knowledge graph as the semantic backbone of the artificial languages.

3.1.1 Graph Sampling Methods

We sampled several high-level synsets from WordNet (Miller, 1995), such as `animal.n.01`, `person.n.01`, `plant.n.02` and `location.n.01`. These roots are used to make the graph cover several major semantic domains and ensure connectivity. To simplify our artificial language, i.e., to eliminate polysemy, we employ synsets rather than lemmas.

Then, we used a Breadth First Search based method to expand the graph with more WordNet noun synsets. We enriched the WordNet backbone with ConceptNet (Speer et al., 2017) by using WordNet lemmas to find corresponding ConceptNet nodes. To make the graph denser, low-degree nodes were enriched with ConceptNet edges when possible.

Additionally, we manually connected some noun nodes with HasQuality edges. These edges enable adjective–noun nominal phrases.

The final graph contains around 2,042 endpoint nodes and 7,181 semantic edges.

3.2 Corpus Generation

3.2.1 Grammar Templates

Structured as a Probabilistic Context-Free Grammar (PCFG), the proposed grammar generates up to 30 grammar templates per *relation* through probabilistic rule combinations. For instance, a triple like (`tiger`, `AtLocation`, `zoo`) can be realized as *tiger populates zoo* or *tiger is situated in zoo*, etc.

3.3 Switch Mechanism

Two languages generated from the same grammar would come out structurally identical. To introduce controlled syntactic differences, we use a switch mechanism, adapted from White and Cotterell (2021). A switch is a binary toggle in $\{0, 1\}$ whose state determines the order of a specific constituent group in every generated sentence.

The VP- and the S-switch control verb–object and subject–verb order, respectively; the NP-switch controls adjective–noun order and remains fixed in the main experiments. We vary these switches to manipulate structure while keeping all other settings identical.

3.3.1 Sentence Generation Pipeline

We start with a triple and sample the grammar template. We then search for modifiers in the form of neighboring adjective nodes in our graph. The noun phrase is expanded in only 70% of cases where possible.

To generate complex and compound sentences, we identify neighboring nodes and edges within the graph, targeting triples that support subject/object coordination or relative clauses. A candidate edge is then sampled to realize either structure. A more detailed description is provided in Appendix A.

This way we have generated a corpus of approximately 50,000 sentences.

3.4 Artificial Languages

We transform the shared generated corpus into two distinct artificial languages by mapping its underlying tokens to two disjoint sets of surface forms.

For each language, we build a dictionary that assigns every concept a single artificial token. The two scripts occupy disjoint Unicode ranges, so the two languages share no tokens at all by construction. For example, the concept `abdomen.n.01` becomes 旅殂 in source language and じぞ in target language. The exact procedure for sampling tokens from each script is given in Appendix B.

4 Evaluation

To compare cross-lingual transfer across configurations, we evaluate model performance using four metrics. We adopt first two metrics from Dufter and Schütze (2020) and adjusted them to our setup. The other two are our own: LM-head prediction and linear probing.

4.1 Word Translation and Sentence Retrieval

Both metrics compare the two languages’ word embeddings directly and use the parallel alignment between them as ground truth, reporting top-1 and top-5 precision.

Word translation: for each content token in source language, we rank all target language content tokens by the cosine similarity of their

embeddings and check whether the aligned token is retrieved.

Sentence retrieval: we represent each sentence as the mean of its word embeddings, and for each source language sentence we rank the target language sentences by cosine similarity and check whether the aligned one is retrieved.

4.2 LM Head

We use the model’s language modelling head directly to test whether a fact has transferred from one language to the other. We hold out a set of facts from target language while keeping them in the source language, and train the model on the resulting corpus. At test time we take a sentence in the target language which never saw the fact, mask its target token, and read the model’s prediction from the language modelling head. Since the model never saw this fact in that language, a correct prediction can only come from cross-lingual transfer.

To obtain a clean measure of transfer, we exclude facts that a model trained on the held-out language (target language) alone can already predict. This is further described in Appendix C. We count a prediction as correct when the model’s highest-scoring token over the full vocabulary is exactly the held-out target token. We also report two looser criteria as a robustness check, and give all three definitions in Appendix D.

4.3 Linear Probing

The LM head only captures transfer that is reflected in the output predictions. A fact may still be aligned across the two languages in the model’s hidden representations without appearing as the top prediction. To look for this, we probe the hidden representations directly.

We reuse the same held-out facts and the same masking. For each probe sentence we freeze the trained model, run it, and take the hidden state at the masked position at every layer. At a given layer, we fit a linear classifier on the representations from the source language that did see the fact, using the tar-

get concept as the label (which is identical across languages), and then apply that same classifier to the representations from the target language that did not. Because the target concept is shared across the two languages, a classifier trained on one side can be tested on the other. The accuracy on the held-out side measures how well a linear probe can predict the fact from its internal representation, even though that language never saw the fact during training.

Probing the internal representations can reveal cross-lingual alignment that is not reflected in the LM-head predictions, and allows us to determine at which layer this alignment becomes detectable. Implementation details are given in Appendix E.

5 Experiments

We train small BERT-like models jointly on both artificial languages using only the MLM objective for all experiments.

5.1 Structural Token Overlap

We aimed to investigate whether sentence structure markers—tokens that appear at fixed, predictable positions—contribute to alignment and transfer in the presence of shared syntactic structure, but in the absence of any lexical overlap.

5.1.1 Setup

The first experiment manipulates the corpus’ only punctuation marks—periods and commas—across three settings: shared, disjoint, and none, to distinguish the effect of shared punctuation from punctuation presence itself.

The special-token experiment asks whether cross-lingual alignment and transfer can be affected by training with shared, language-specific, or absent BERT’s [CLS] and [SEP] tokens, being sentence boundary markers on model-input level.

5.1.2 Results

Comprehensive results for this experiment can be found in Appendix F.1

We find that Word Translation and Sentence Retrieval precision for language-specific punctuation is significantly¹ lower than for shared or absent. For [CLS]/[SEP], only Word Translation shows a stable effect of disjoint.

The linear probe shows the same pattern for punctuation: poorer alignment under disjoint punctuation corresponds to less linearly recoverable transfer. However, the pattern does not reach the model’s final predictions.

There is no evidence that training with [CLS]/[SEP] helps transferred information to become internally accessible. LM-head transfer was similarly low and indistinguishable across settings.

Overall, there is no sign that shared punctuation or [CLS]/[SEP] act as additional anchor points in our setting. Language-specific punctuation appears to have hurt alignment. Sharing or differentiating [CLS]/[SEP] during training may have a small effect on lexical alignment, but does not influence sentence-level alignment or knowledge transfer.

5.2 Lexical Overlap

In this experiment, we investigate whether the degree of lexical overlap influences cross-lingual alignment of two artificial languages.

5.2.1 Setup

We test target corpus overlaps of approx. 0, 2.5, 5, 7.5, and 10% using tokens from three frequency pools: *high-frequency* (ranks 1–10), *mid-frequency* (11–50), and *low-frequency* (51–200). Tokens are randomly selected from a pool, shifted to a common character set, and accumulated until reaching the target percentage. This yields 13 distinct configurations across 15 total strategy–percentage combinations.

5.2.2 Results

In Word Translation, adding lexical overlap significantly improves performance over the 0% baseline, despite percentage-wise fluctuations without monotonic trends². In Sen-

tence Retrieval, these gains are less pronounced. The low-frequency strategy yields the strongest improvements across both metrics, significantly outperforming equivalent target percentages.

In Linear Probing, overlap percentages above 0% consistently outperform the baseline, best visible in the final layer. The low-frequency strategy yields the strongest improvements.

LM-head strict top-1 accuracies remain consistently below 5% across all conditions. Increasing lexical overlap yields no significant performance gains.

Lexical overlap facilitates cross-lingual alignment, primarily taking place within the model’s internal representations. Aligning a diverse set of content words yields stronger alignment than aligning high-frequency function words.

For more comprehensive evaluation results, see Appendix F.2.1.

5.3 Anchor Necessity

5.3.1 Setup

The observed transfer even between languages with no shared tokens suggests that the models rely on structural similarities rather than overlap. We therefore gradually reduce their structural similarity, using the switch mechanism to control the word order of target language from identical to increasingly different. Source language is fixed at 111 while target language ranges over 000, 001, 011, 111.³

5.3.2 Results

Word translation, sentence retrieval, and linear probing all follow the same pattern. Transfer stays at floor for 0%, 33%, and 67% similarity, and appears only at 100%, when the two languages have identical word order. Thus,

and investigates: (a) whether a condition significantly outperforms the 0% baseline, and (b) whether the low-frequency setting significantly outperforms other settings at an equivalent target percentage.

³We quantify structural similarity as the fraction of switch settings the two languages have in common, i.e., one minus the normalised Hamming distance between their switch configurations. Source language is fixed at 111 and target language ranges over 000, 001, 011, 111, giving 0, 33, 67, 100% similarity.

¹For significance testing details, see Appendix G.

²Since no monotonic trends were observed, significance testing is based on pairwise comparisons (Fisher’s exact test)

shared tokens are not necessary for transfer, but identical structure is in this setting.

The LM head, by contrast, shows no corresponding effect. Its strict top-1 accuracy stays around 7% across all similarity levels, including identical word order. The same pattern holds under the more lenient criteria. The increase at identical structure is significant for the probing results, while no corresponding effect is found for the LM-head results. This indicates that cross-lingual alignment can be detected in the model’s hidden representations even when it is not reflected in the LM-head predictions.

For more comprehensive evaluation results, see Appendix F.3.1.

5.4 Graph Density

In this experiment, we investigate whether graph density influences cross-lingual alignment. Defined as the proportion of retained edges from the full graph, higher density provides each concept with more semantic neighbors and more relational contexts, meaning the languages are more structurally complete and connected.

5.4.1 Setup

All graph-density conditions are sampled from the same full graph: low (40%), medium (60%), high (80%), and full (100%).

5.4.2 Sampling Strategies

To ensure that all conditions except graph density were kept similar, we used different sampling methods:

- *Coverage-first sampling* keeps all nodes connected as much as possible.
- *Relation-matched sampling* tries to keep the relation distribution close to the full graph.

The experiment also compares downsampled training and full-corpus training. In downsampled training, the number of training sentences is the same across all densities. In full-corpus training, the model can use all generated sentences from the condition.

5.4.3 Results

For word translation metric, denser graphs generally lead to a significantly better cross-lingual alignment (under the Spearman permutation test), whereas for sentence retrieval metric full graph performs slightly worse than 80% density, which could be the result of insufficient runs per condition.

The cross-lingual alignment changes more clearly than the monolingual MLM score, so the density effect is more related to alignment. The MLM score even drops as the graph becomes denser. This fall is within expectations because the downsampling leads to relatively less sufficient model training for denser graphs.

For graph density, the relation-based LM-head results show a slight, but significant upward pattern. In contrast, the linear probe shows a much stronger increase over the same relation-based density sweep. Thus, denser relational structure appears to make cross-lingual information more recoverable in the hidden representations, while the model’s native LM head expresses this signal only weakly as direct token prediction.

For more comprehensive evaluation results, see Appendix F.4.1.

5.5 Semantic Overlap

In this experiment, we investigate whether languages that shares more semantic information align better. We define semantic overlap as the percentage of graph triples shared between a source and a target language. The total number of edges for each language is fixed, but the number of shared edges changes. Therefore, this experiment focuses on the effect of shared semantic facts.

5.5.1 Setup

The sampling is stratified by relation and source type. This keeps the relation distribution controlled across overlap conditions. We used a sub-graph of 2870 edges instead.

5.5.2 Results

Semantic overlap strongly improves cross-lingual alignment. Word translation precision

significantly increases as semantic overlap increases, and so does the sentence retrieval. At 0% overlap, word translation is close to the bottom, and sentence retrieval is not available because there are no shared triples for parallel sentence retrieval.

The LM-head results show a positive pattern. The overall Spearman trend, however, is not statistically significant, so the ordered trend should be interpreted cautiously. Nonetheless, the endpoint comparison between 75% and 0% overlap is significant under Fisher’s exact test, indicating that the highest-overlap condition performs reliably better than the no-overlap condition. This effect is still much weaker than the linear probe result.

Together, these results suggest that semantic overlap strongly improves the recoverability of held-out information in the hidden representations, while the LM head shows only a smaller and less monotonic improvement.

For more comprehensive evaluation results, see Appendix F.5.1.

6 Conclusion

This project studies cross-lingual transfer in a controlled artificial-language setting. We built two artificial languages from a shared knowledge graph and generated sentences with PCFG, conducted several experiments to test the dependency of this transfer. The results suggest that cross-lingual transfer is supported by a combination of lexical, structural, and semantic signals, rather than by anchor tokens alone.

Our evaluation further shows that models are able to internally align and transfer information across languages, even when this is not observable in the output.

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Limitations and Lessons Learned

In this project, we tried to conduct a small research and to consider all the different methodological decisions this requires. Yet not every aspect of our project meets full standards of an actual scientific study.

We cannot yet state that our findings can be transferred to natural language corpora. Our sentence structure is relatively simple, and the semantic relations coming from the knowledge graph may not fully reflect those found in natural languages. For example, `HasQuality` edges were introduced manually and, despite careful consideration, may introduce some subjective bias.

Given the number of experiments and models that had to be trained, we unfortunately could not run every condition with 5–10 random seeds. Some of our findings we therefore should interpret with caution.

The projects we had worked on before mainly focused on answering one question, while here we had several sub-questions in one shared research context. We had to make many careful decisions to maintain consistency across experiments and settings, and to ensure that our results remained interpretable and comparable. Designing the toy corpus itself, as well as evaluation metrics that matched our research questions, took a great deal of planning and consideration.

Throughout the project, we met as a group weekly to discuss methodology and progress and to distribute the workload. We accommodated everyone’s schedules when planning meetings and assigning tasks. At the same time, we remained available to each other throughout the week for discussions and troubleshooting. We allowed each team member to take ownership of their tasks according to interests, while still maintaining a general overview of the project. We gained good practical experience in coordinating our work through GitHub and presenting it together.

Declaration of AI Usage

Generative AI tools, such as ChatGPT and Claude Code, were used selectively as auxiliary tools. Their use primarily included troubleshooting and debugging programming-related issues, clarifying technical concepts, and supporting the reading and understanding of relevant academic literature.

During our project, AI tools were used for:

- Helping understand papers.
- Assisting with coding (e.g., reducing repetitive writing).
- Assisting with code debugging and formatting.
- Organizing data (e.g., plotting).
- Assisting with documentation.
- Assisting with literature search and information retrieval.

During our project, AI tools were **not** used for:

- Project planning and management.
- Selecting algorithms and methods (e.g., knowledge graph generation, PCFG implementation).
- Reviewing and verifying critical code components.
- Designing experimental protocols.
- Analyzing and interpreting experimental results.
- Writing technical protocols and the final report.

We remain fully responsible for the accuracy, integrity, and final content of the submitted work.

A Sentence Generation Pipeline

We start the pipeline with a triple and two constraints: `n_simple` and `n_sample`. First, we sample the verb—or more precisely, the grammar template. Next, we look at our source and target entities. These entities can receive a modifier in the form of an adjective, depending on whether they are connected to a neighboring adjective node in our graph. Even when a matching adjective is available, our rules dictate that the noun phrase will expand into an Adjective + Noun structure in only 70% of cases, while the remaining 30% will remain a bare Noun. In 30% of the cases we also generate minimal pair sentences of our generated sentence, differing only in an adjective: *red tiger occupies zoo* and *rare tiger occupies zoo*. These sentences do not use up the constraint limits.

At this stage, we have a complete, simple sentence ready. We then increment both the `n_simple` and `n_sample` counters by 1, and sample another template. We repeat this process until we have successfully generated 5 diverse simple sentences.

Since simple sentences alone do not yet allow for enough structural diversity, in the next step we want to generate complex and compound sentences from our triples. To achieve this, we look for neighboring nodes and edges in our graph. Specifically, we target three types of triples:

1. *Same-relation*: Triples sharing both the same source and the same relation, which enables target coordination.
2. *Same-target*: Triples sharing the same relation and target, enabling source coordination.
3. *Expansion*: Triples where a relation is defined as a possible extension, which allows us to embed relative clauses—for example: *tiger is animal, that lives in zoo*.

After identifying these sibling edges, we sample them one at a time to generate either a coordination or a relative clause. We again check for potential adjective modifiers. Once the final sentence is emitted, we increment the `n_sample` counter by 1. Since we have already generated 5 simple sentences, our goal here is to generate 3 additional complex ones.

B Artificial Token Generation

The artificial token generation process is simple. The goal is to replace each real concept with a new artificial surface form.

First, the program chooses a Unicode character set. For example, if the `cjk` preset is used, the characters are sampled from the CJK Unicode range. If the `hiragana` preset is used, the characters are sampled from the Hiragana Unicode range. Other presets, such as `cyrillic` and `yi`, work in the same way.

Then, for each real lemma, the program randomly decides the length of the artificial token. The default average length is about three characters. The final length is limited to a fixed range, usually between one and eight characters.

After the length is chosen, the program randomly samples characters from the selected Unicode character set. These characters are joined together to form one artificial token. For example, a concept such as `animal.n.01` may be mapped to an artificial token such as 你而财洵.

The program also checks that each artificial token is unique. If a newly generated token has already been used for another concept, the program generates a new one until it finds an unused token.

C Probe Fact Selection

A probe is a single fact that we hold out from one language during training and query at test time. Not every fact makes a valid probe, so we select them with a few rules. We mask the target entity of a fact (a, r, b) : the model is given a r [MASK] and has to predict b . A fact is a usable

probe only if:

- the target is unique given (a, r) : there is no other fact (a, r, b') with $b' \neq b$, so the masked answer is determined rather than one of several equally valid fillers;
- the target is grounded: $\deg(b) \geq 2$, so the held-out entity appears in at least one other fact and the model had a chance to learn it from a different context.

Among the facts that pass these rules, we keep a relation only if it has at least 100 facts in total, so that holding out a fixed number still leaves enough of the relation to be learned, and we then sample 50 probes per kept relation (seed 42).

Monolingual reason filter. A held-out fact is a fair test of transfer only if the model cannot already answer it without having seen it. Some targets are recoverable from the sentence alone. Take the fact $(tiger, \text{hyponym}, animal)$, realised as $tiger_B \text{ is a kind of } [MASK]$: a model can fill in $animal$ from the relation and the subject, without ever having learned this specific fact, so a correct prediction here would not show transfer. To catch such cases, we train a model on the held-out language alone and remove every candidate whose masked target this monolingual model already predicts as its top choice. A fact like $(tiger, \text{AtLocation}, zoo)$ in $tiger_B \text{ populates } [MASK]$ has no such shortcut, so it is kept.

Masking the relation. We also tried a second track that masks the relation instead of the entity, giving the model $a \text{ } [MASK] \text{ } b$ and asking it to predict r . This did not work. A relation is largely fixed by its two entities, so the monolingual model already predicted it for almost every candidate, and too few probes survived the reason filter to give a reliable estimate. We therefore report only the mask-entity track.

D LM Head Evaluation

We report three correctness criteria, from loose to strict. Continuing the example $tiger_B \text{ populates } [MASK]$ with held-out target zoo :

- lenient: ranking over Language B tokens only, the top token is zoo_B ;
- strict_concept: ranking over both vocabularies, the top token denotes the concept zoo (either zoo_A or zoo_B counts);
- strict_token: ranking over both vocabularies, the top token is exactly zoo_B .

Each is reported at top-1 and top-5. strict_token is our primary criterion; the other two show whether the conclusion holds under a looser requirement.

E Linear Probing

We probe the held-out facts with the same masking used above. For each probe sentence, illustrated with $tiger_B \text{ populates } [MASK]$:

- freeze the trained model and run it, taking the hidden state at the $[MASK]$ position at every layer (one vector per layer);
- at each layer, fit a linear classifier on the Language A vectors with the target concept (zoo) as the label;
- apply that same classifier to the Language B vectors of the same facts.

The accuracy on Language B at each layer measures how much of the fact is linearly readable from its internal representation, even though Language B never saw it in training. Figure 1 shows the full pipeline.

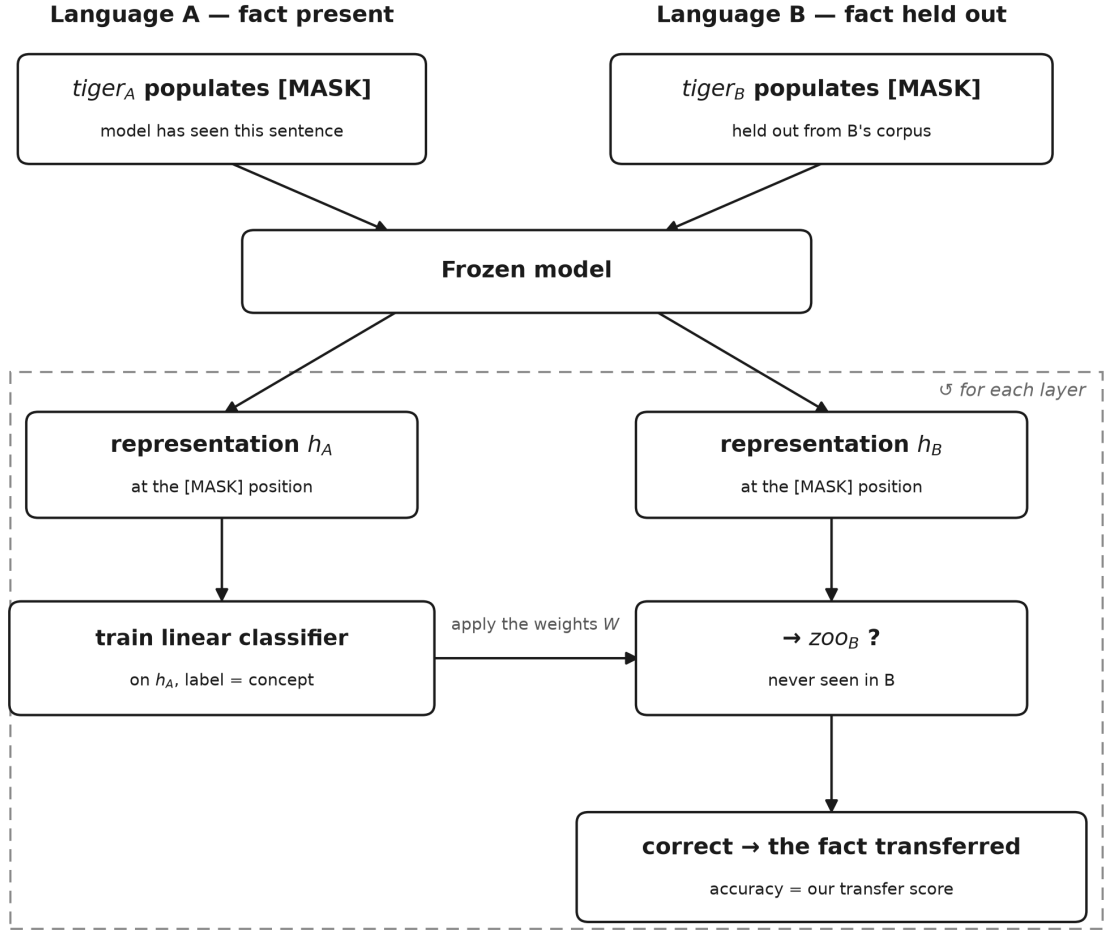


Figure 1: Linear probing pipeline.

F Additional Experimental Results

F.1 Structural Tokens

F.1.1 Limitations and Remarks

For the structural tokens experiments, our models were trained with three seeds. We can observe sensitivity to random seed in several conditions, particularly for Sentence Retrieval. Linear probe and LM-head results for a single seed, which is why we especially can not treat the results of these metrics as definitive.

Another remark we would like to make is that punct-shared and st-none, despite sharing corpus settings, are not directly comparable. In each experiment, only the marker investigated is not getting masked by MLM, which means that period and comma can not be masked in punct-shared and can in st-none.

We hypothesise that in our setting, since sentence boundaries are already implicit in the structure of each training sequence, and period is present, shared [CLS] and [SEP] provide little additional information. As for punctuation, we suggest that the period in our punct-disjoint setting may act as language-identifying signal, giving the model a reason to keep their representations separate. The comma is less frequent, so it likely acts as a weaker version of the same signal.

F.1.2 WT and SR results

Tab.1 summarises the effect on alignment. The setting punct-none, containing neither punctuation nor boundary special tokens, can be considered a baseline with no sentence structure markers.

Exp.	Metric	Shared	None	Disjoint
PUNCT	WT	0.74 ± 0.01	0.75 ± 0.03^4	0.46 ± 0.08
	SR	0.83 ± 0.06	0.83 ± 0.06	0.49 ± 0.09
ST	WT	0.79 ± 0.03	0.80 ± 0.03	0.69 ± 0.03
	SR	0.83 ± 0.06	0.87 ± 0.04	0.82 ± 0.05

Table 1: Mean top-1 word translation (WT) and sentence retrieval (SR) precision over 3 seeds, reported as mean $\pm$ standard deviation.

F.1.3 Significance of the WT and SR Results

The main section reports that the disadvantage of disjoint special tokens is consistent only for Word Translation. We additionally report our significance tests results to illustrate this, as well as another trend: although the mean shows none setting mostly outperforming shared for both punctuation and special tokens, this difference was not consistently significant, with the direction varying across seeds.

In the following Tab. 2, significant results ($p < .05$) are shown in bold. The details of our significance test strategy for all experiments can be found in Appendix G.

Experiment	Metric	Comparison	Seed 42	Seed 43	Seed 44
PUNCT	WT	shared > disjoint	6.75×10^{-126}	4.78×10^{-51}	1.55×10^{-57}
		none > disjoint	1.77×10^{-147}	2.54×10^{-64}	5.82×10^{-42}
		shared vs. none	0.040	0.056	0.015
	SR	shared > disjoint	6.96×10^{-35}	1.61×10^{-40}	5.54×10^{-22}
		none > disjoint	5.23×10^{-68}	7.70×10^{-21}	1.05×10^{-16}
		shared vs. none	4.88×10^{-8}	3.76×10^{-5}	0.189
ST	WT	shared > disjoint	4.47×10^{-22}	0.0067	4.41×10^{-19}
		none > disjoint	8.57×10^{-15}	3.84×10^{-5}	5.53×10^{-32}
		shared vs. none	0.058	0.148	0.0043
	SR	shared > disjoint	0.375	0.9999	7.57×10^{-7}
		none > disjoint	0.012	0.947	4.82×10^{-9}
		shared vs. none	0.063	0.043	0.397

Table 2: Fisher’s exact test results for word translation (WT) and sentence retrieval (SR) in the punctuation and special-token experiments. Shared > disjoint and none > disjoint tested one-sided, shared vs. none tested two-sided.

F.1.4 Sentence Retrieval for Punctuation

We noticed that since [CLS] and [SEP] are not included in the static embedding based sentence representation, while period and comma are, the punctuation’s and special tokens’ evaluations may not be strictly equivalent. Thus, we tried additionally recomputing punctuation Sentence Retrieval Precision with period and comma excluded from the sentence mean at evaluation time. The metric otherwise remains unchanged. The same overall pattern can be seen: disjoint performed worse than shared or none (see Tab. 3).

Experiment	Metric	Shared	None	Disjoint
PUNCT	Standard SR	0.83 ± 0.06	0.78 ± 0.03	0.49 ± 0.09
	Punctuation excluded	0.86 ± 0.05	0.82 ± 0.08	0.55 ± 0.08
ST	Standard SR	0.83 ± 0.06	0.83 ± 0.06	0.82 ± 0.05

Table 3: Mean top-1 Sentence Retrieval precision over 3 seeds, mean $\pm$ SD.

F.1.5 Linear Probing and LM Head

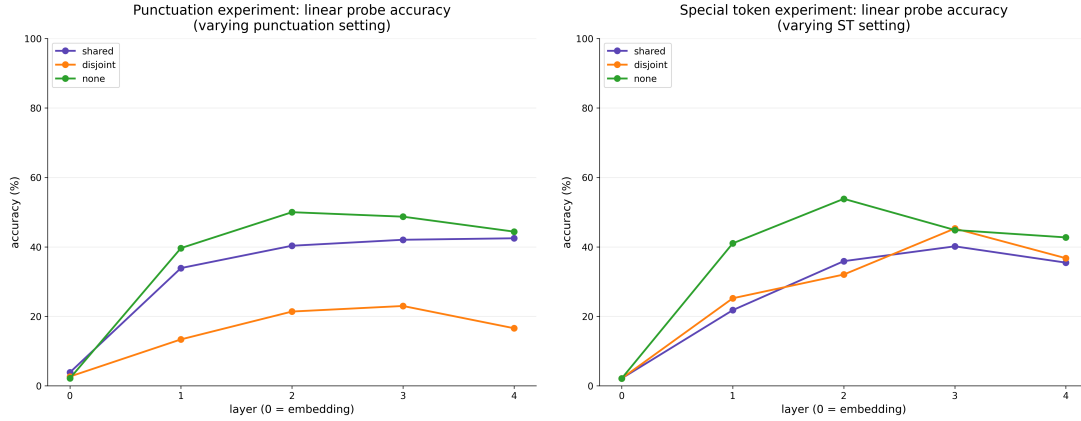


Figure 2: Layer-wise probing accuracy for punctuation (left) and ST (right), single seed. The differences between settings were significant only for punctuation comparisons against the disjoint.

Experiment	Shared	None	Disjoint
PUNCT	0.0343	0.0388	0.0214
ST	0.0556	0.0342	0.0513

Table 4: Strict-token top-1 LM-head accuracy for the punctuation (PUNCT) and special-token (ST) experiments, single seed. None of the comparisons reached significance.

F.2 Lexical Overlap

F.2.1 Comprehensive Evaluation Results

Table 6: Comprehensive evaluation results for the lexical overlap experiment (seed 42).

(a) Word Translation and Sentence Retrieval precision.

Condition	Word Top-1	Word Top-5	Sentence Top-1	Sentence Top-5
high_P0	0.7645	0.8945	0.8347	0.9613
high_P2	0.7630	0.8985	0.8247	0.9447
high_P5	0.7880	0.9135	0.8200	0.9527
high_P7	0.7955	0.9265	0.8220	0.9527
high_P10	0.7840	0.9170	0.8460	0.9620
mid_P0	0.7645	0.8945	0.8347	0.9613
mid_P2	0.8240	0.9395	0.8047	0.9380
mid_P5	0.7905	0.9220	0.0041	0.9620
mid_P7	0.8195	0.9305	0.8847	0.9613
mid_P10	0.8065	0.9255	0.8307	0.9567
low_P0	0.7645	0.8945	0.8347	0.9613
low_P2	0.8381	0.9489	0.8880	0.9733
low_P5	0.8288	0.9411	0.0146	0.9467
low_P7	0.8433	0.9357	0.9300	0.9893
low_P10	0.8770	0.9538	0.9047	0.9807

(b) Layer-wise probing accuracy.

Condition	L0	L1	L2	L3	L4
high_P0	0.0324	0.2842	0.4460	0.4496	0.3813
high_P2	0.0324	0.4460	0.4281	0.4856	0.5000
high_P5	0.0324	0.3489	0.4173	0.4496	0.4317
high_P7	0.0324	0.4281	0.5432	0.5180	0.5396
high_P10	0.0324	0.3165	0.4640	0.4496	0.4245
mid_P0	0.0324	0.2842	0.4460	0.4496	0.3813
mid_P2	0.0324	0.4281	0.4173	0.5252	0.4532
mid_P5	0.0324	0.3453	0.4209	0.4065	0.3597
mid_P7	0.0324	0.4532	0.5576	0.5288	0.5252
mid_P10	0.0324	0.4892	0.5144	0.4748	0.4820
low_P0	0.0324	0.2842	0.4460	0.4496	0.3813
low_P2	0.0324	0.4964	0.5216	0.4820	0.4820
low_P5	0.0324	0.4964	0.6259	0.5755	0.5432
low_P7	0.0324	0.5036	0.4784	0.5108	0.5432
low_P10	0.0324	0.5144	0.6547	0.6115	0.6007

(c) LM Head accuracy.

Condition	Strict Tok.		Strict Conc.		Lenient	
	T-1	T-3	T-1	T-3	T-1	T-3
high_P0	0.0360	0.0863	0.0360	0.0899	0.0360	0.0863
high_P2	0.0180	0.0827	0.0180	0.0863	0.0180	0.0827
high_P5	0.0252	0.0576	0.0288	0.0647	0.0252	0.0647
high_P7	0.0468	0.0827	0.0468	0.0935	0.0468	0.0863
high_P10	0.0468	0.1043	0.0468	0.1079	0.0468	0.1043
mid_P0	0.0360	0.0863	0.0360	0.0899	0.0360	0.0863
mid_P2	0.0432	0.0827	0.0468	0.0935	0.0432	0.0827
mid_P5	0.0432	0.0935	0.0468	0.0971	0.0432	0.0935
mid_P7	0.0288	0.0827	0.0288	0.0827	0.0288	0.0827
mid_P10	0.0216	0.0827	0.0216	0.0827	0.0216	0.0827
low_P0	0.0360	0.0863	0.0360	0.0899	0.0360	0.0863
low_P2	0.0576	0.0863	0.0576	0.0899	0.0576	0.0863
low_P5	0.0252	0.0540	0.0324	0.0683	0.0252	0.0576
low_P7	0.0324	0.0719	0.0396	0.0899	0.0324	0.0719
low_P10	0.0288	0.0647	0.0288	0.0647	0.0288	0.0755

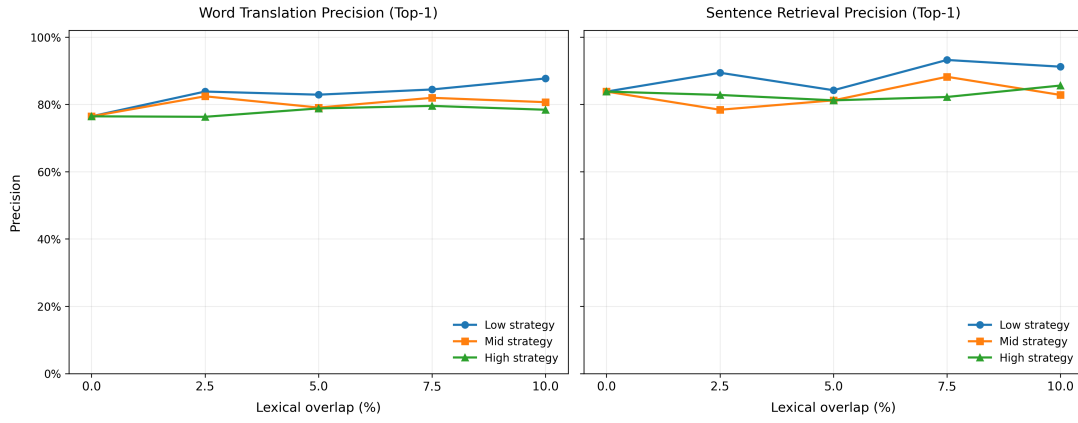


Figure 3: Top-1 precision for word translation (left) and sentence retrieval (right) across varying levels of lexical overlap under three frequency strategies.

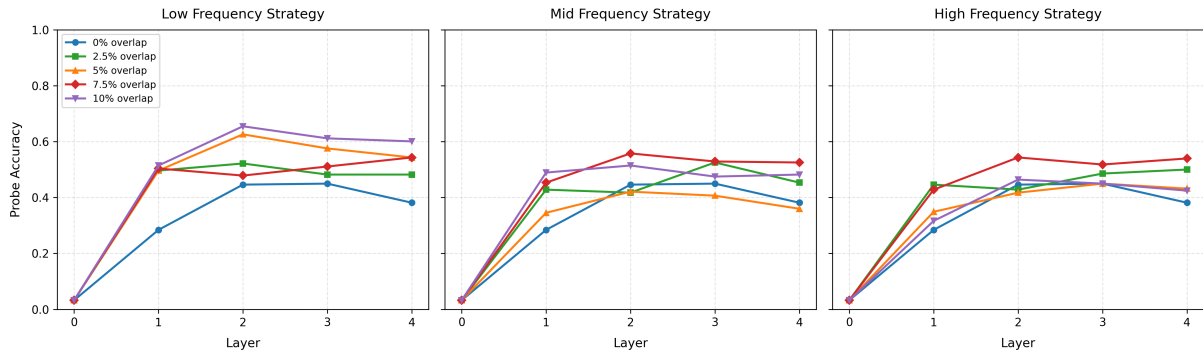


Figure 4: Layer-wise probing accuracy for the low (left), mid (middle), and high (right) sampling strategies across varying degrees of lexical overlap.

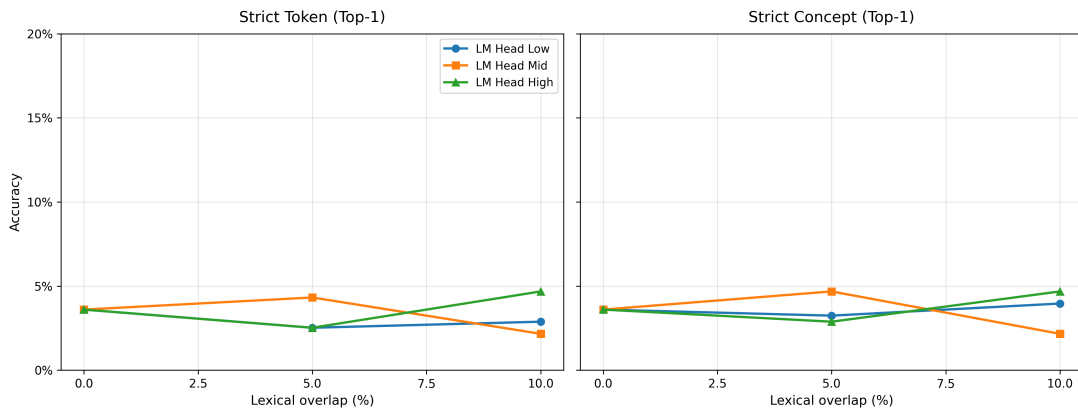


Figure 5: strict_token (left) and strict_concept (right) top-1 LM Head accuracy across varying levels of lexical overlap under three frequency strategies.

F.3 Anchor Necessity

F.3.1 Comprehensive Evaluation Results

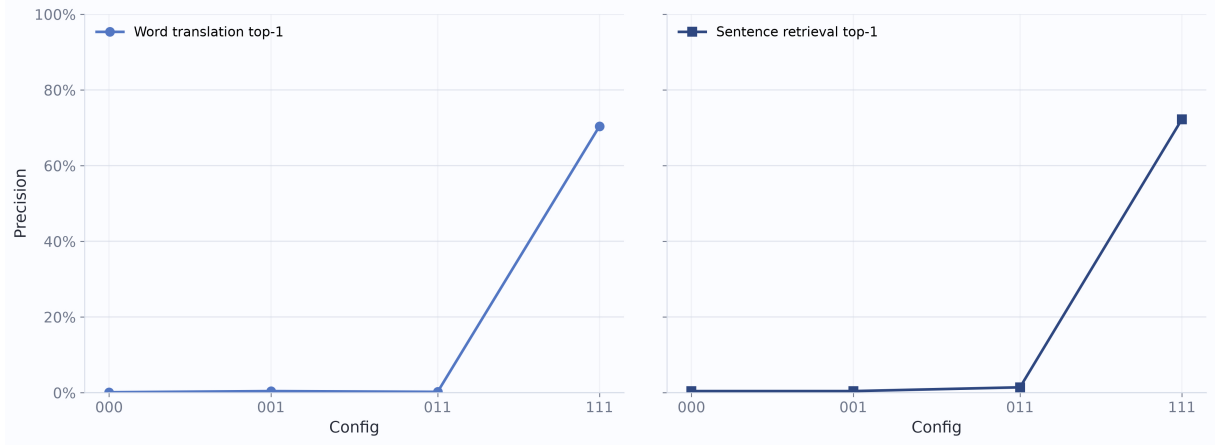


Figure 6: Word translation and sentence retrieval top-1 precision under zero shared tokens, as structural similarity varies.

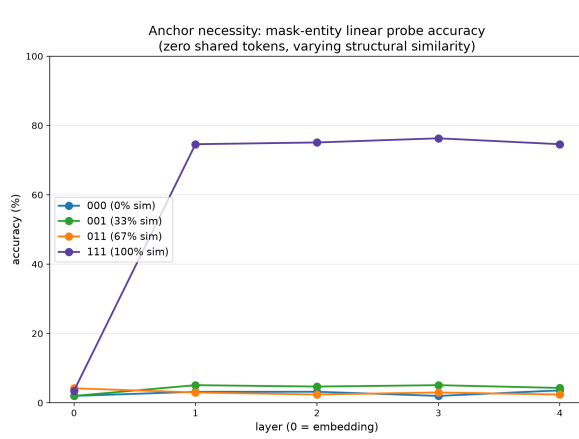


Figure 7: Linear probing accuracy per layer.

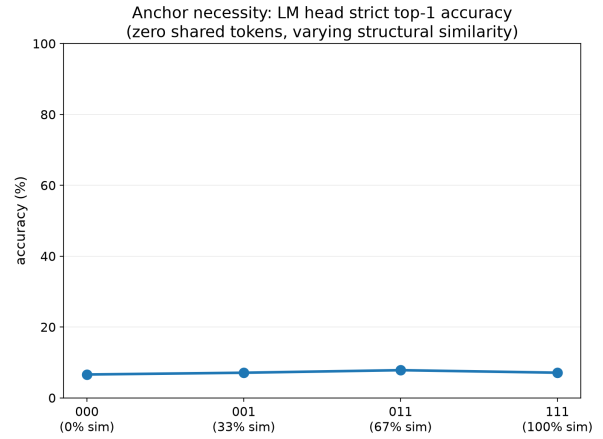


Figure 8: LM head strict top-1 accuracy.

F.4 Graph Density

F.4.1 Comprehensive Evaluation Results

The experimental results in this section support the conclusions presented in the main text.

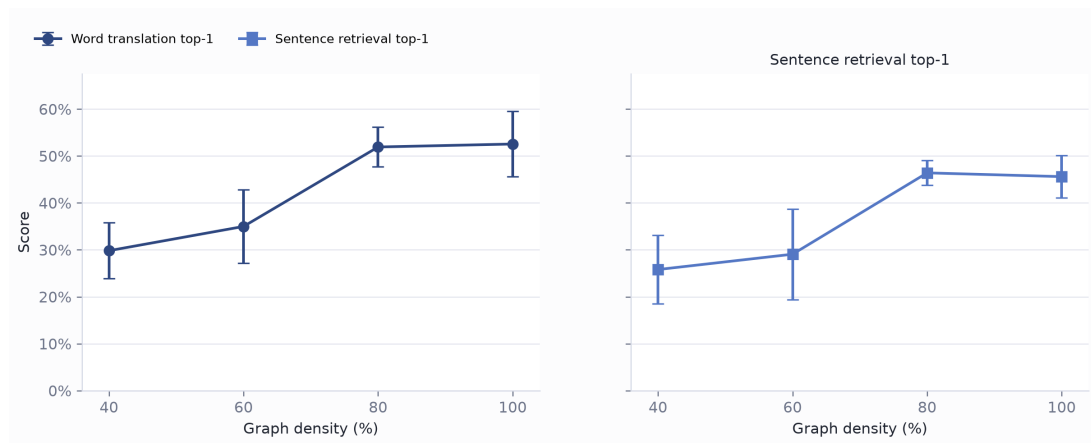


Figure 9: Results of graph density multilingual alignment. Left: word translation metric, right: sentence retrieval metric. Downsampling and relation-based sampling strategies are shown here. A more comprehensive experimental results in Appendix F.4.2.

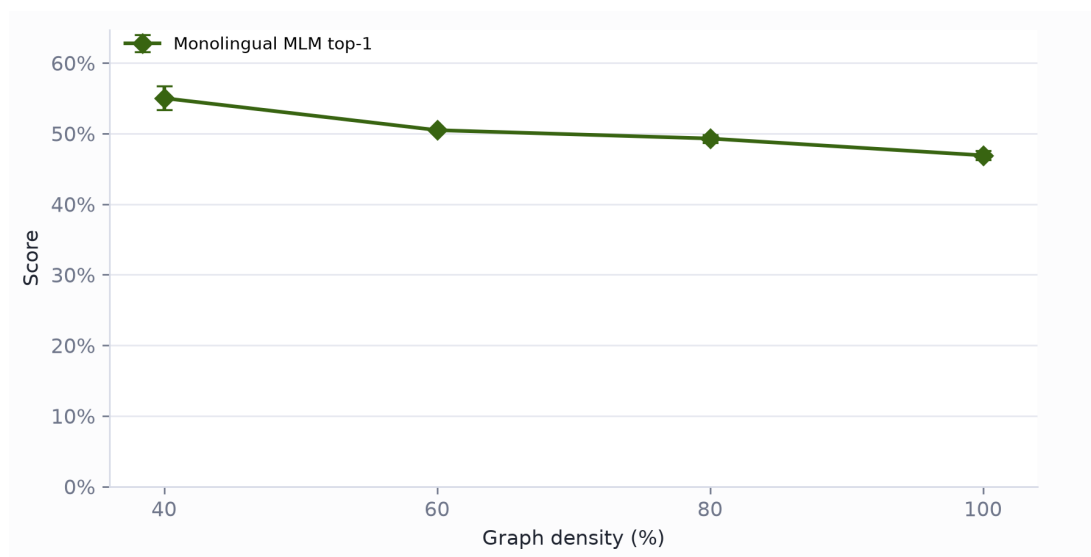


Figure 10: Result of graph density monolingual MLM. Downsampling and relation-based sampling strategies are shown here. A more comprehensive experimental results in Appendix F.4.2.

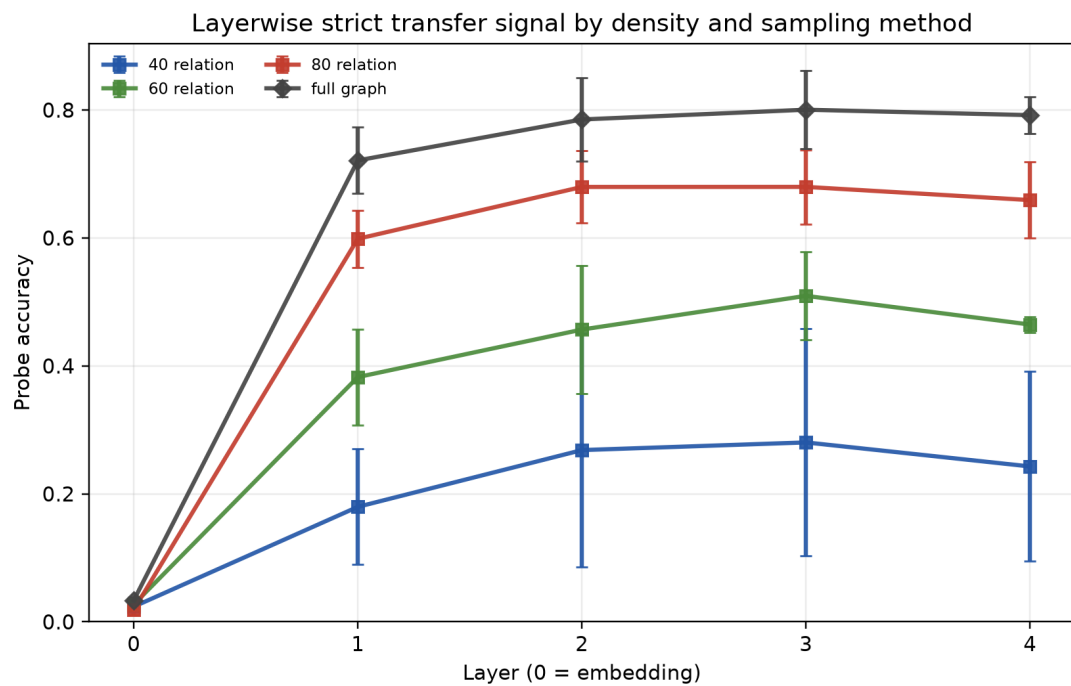


Figure 11: Linear probe results for graph density. Results of relation-based sampling strategy are shown here.

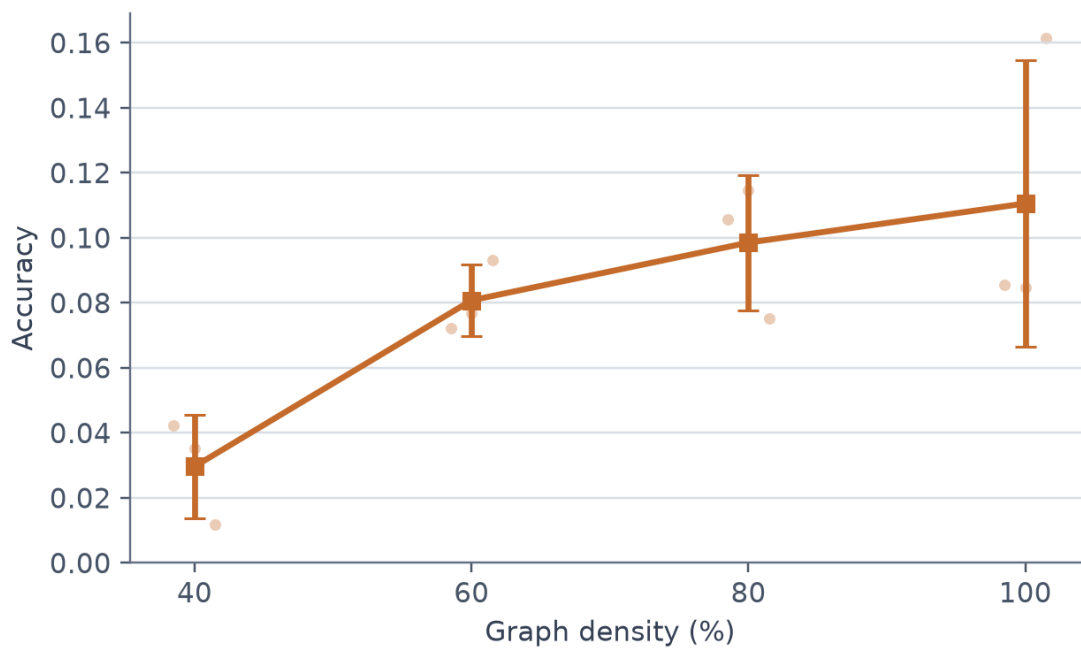


Figure 12: LM head result for graph density. Results of relation-based sampling strategy are shown here. A more comprehensive experimental results in Appendix F.4.2.

F.4.2 A Comparison of Sampling Methods

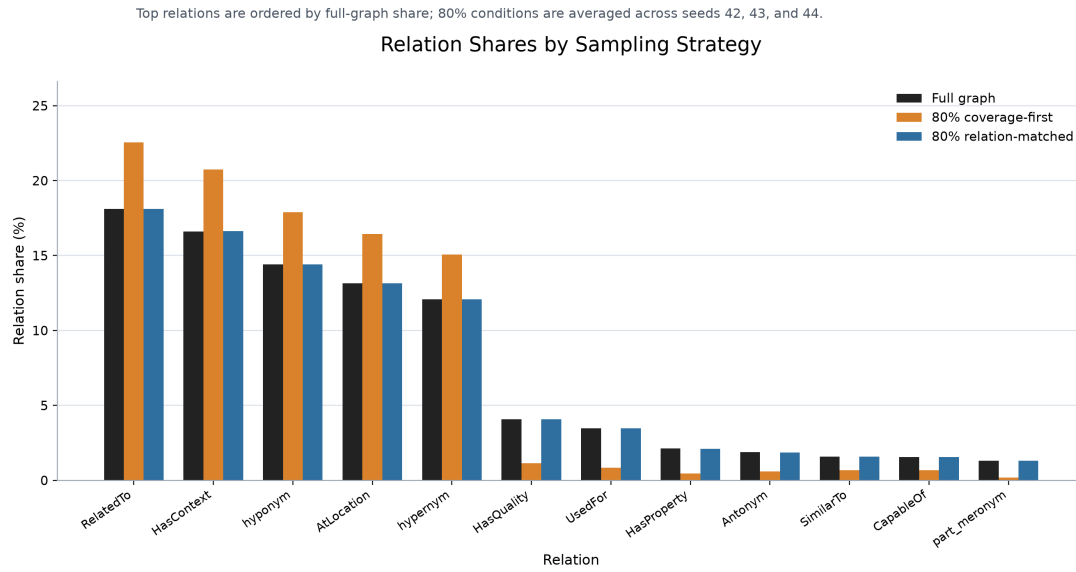


Figure 13: Relation distribution comparison. Only Top-12 relations are shown.

As shown in Figure 13. The relation distribution of graphs sampled with relation-based method is similar to that of the full graph.

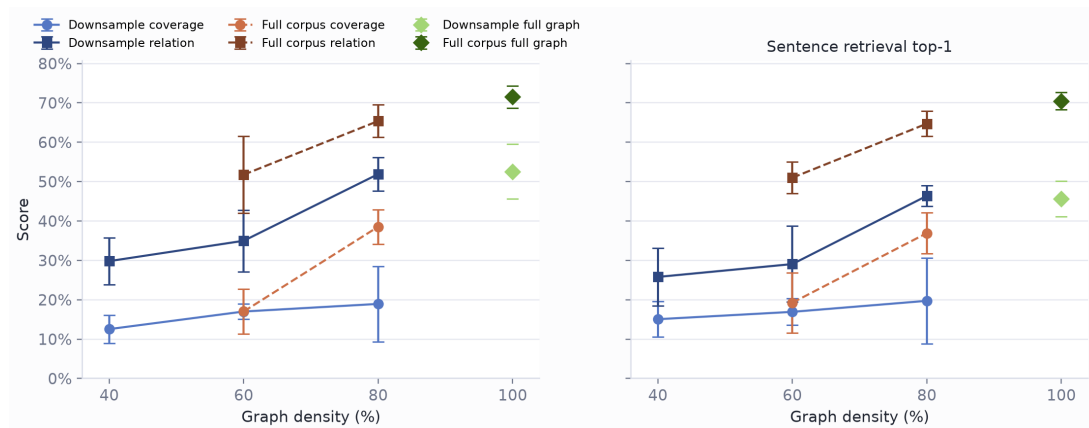


Figure 14: Results of graph density multilingual alignment. Left: word translation metric, right: sentence retrieval metric.

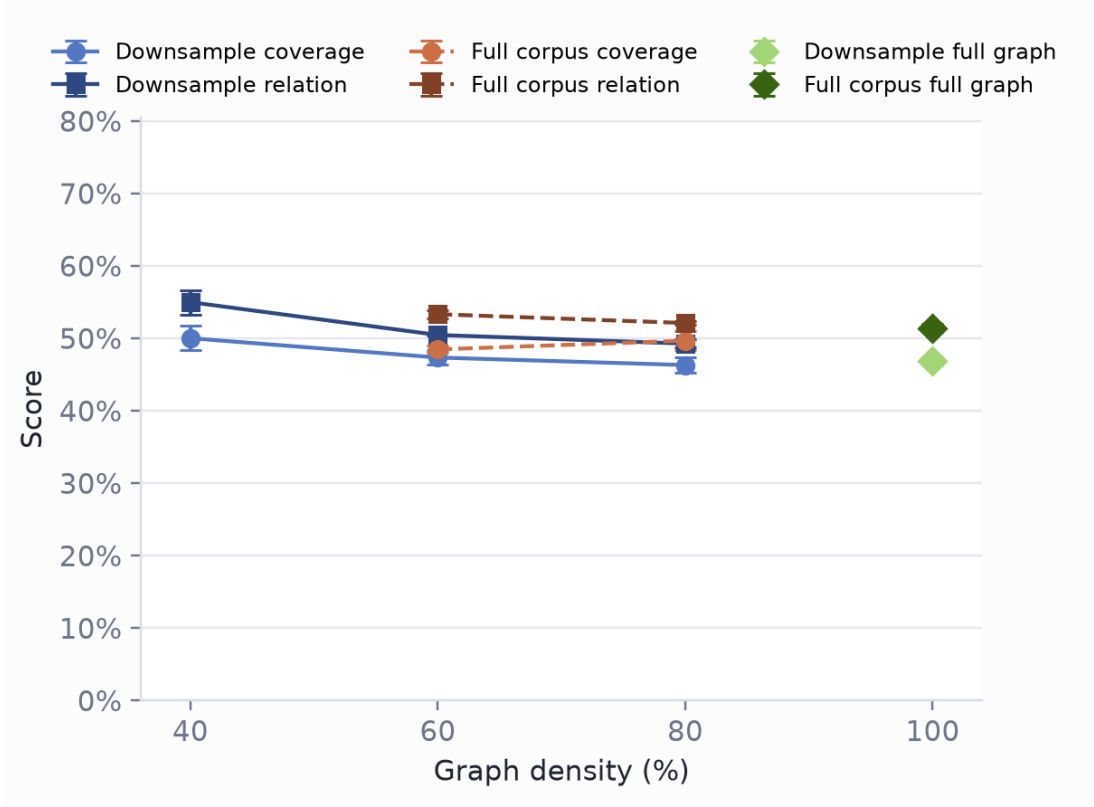


Figure 15: Result of graph density monolingual MLM.

LM-head Accuracy Across Graph Density

strict-token criterion; points are seeds, lines are seed means, error bars are ± 1 SD.

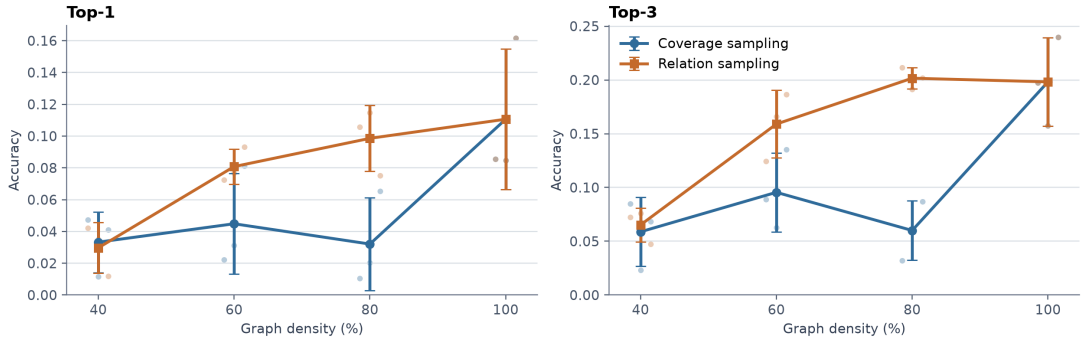


Figure 16: LM head comprehensive results for graph density.

As shown in Figure 14 and Figure 15, the relation-matched strategy performs better than the coverage-first strategy. This suggests that not only the number of edges matters, but also the balance of relation types. If the graph is dense but the relation distribution is distorted, the model may receive a less stable semantic signal.

Full-corpus training is also stronger than downsampled training in most density settings. This is especially clear in the higher-density conditions. A possible explanation is that denser graphs create more sentence variants, and full-corpus training lets the model see more of this semantic evidence.

For LM head, coverage-first sampling also shows an upward trend. However, the accuracy drops at the 80% condition. We cannot find a clear explanation for this change, which could be some unknown noise.

Condition	Shared edges	L1-only edges	L2-only edges	Total per language
0%	0	2,870	2,870	2,870
25%	718	2,152	2,152	2,870
50%	1,435	1,435	1,435	2,870
75%	2,152	718	718	2,870
100%	2,870	0	0	2,870

Table 7: Semantic overlap settings.

F.5 Semantic Overlap

F.5.1 Comprehensive Evaluation Results

The experimental results in this section support the conclusions presented in the main text.

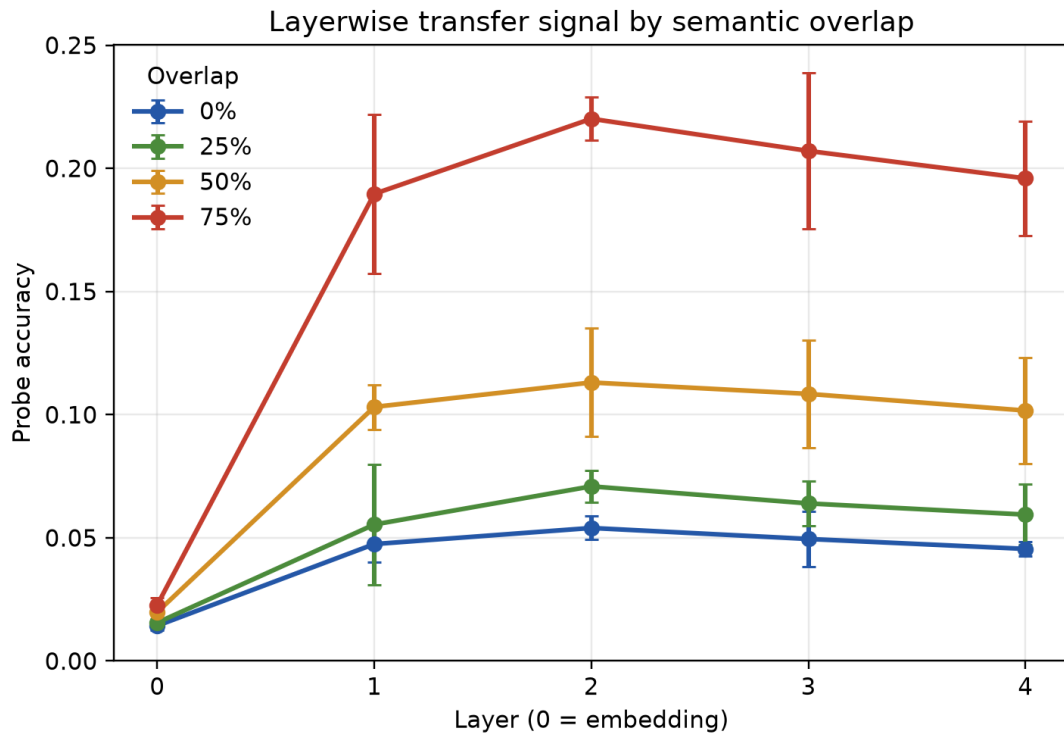


Figure 18: Linear probe results for semantic overlap.

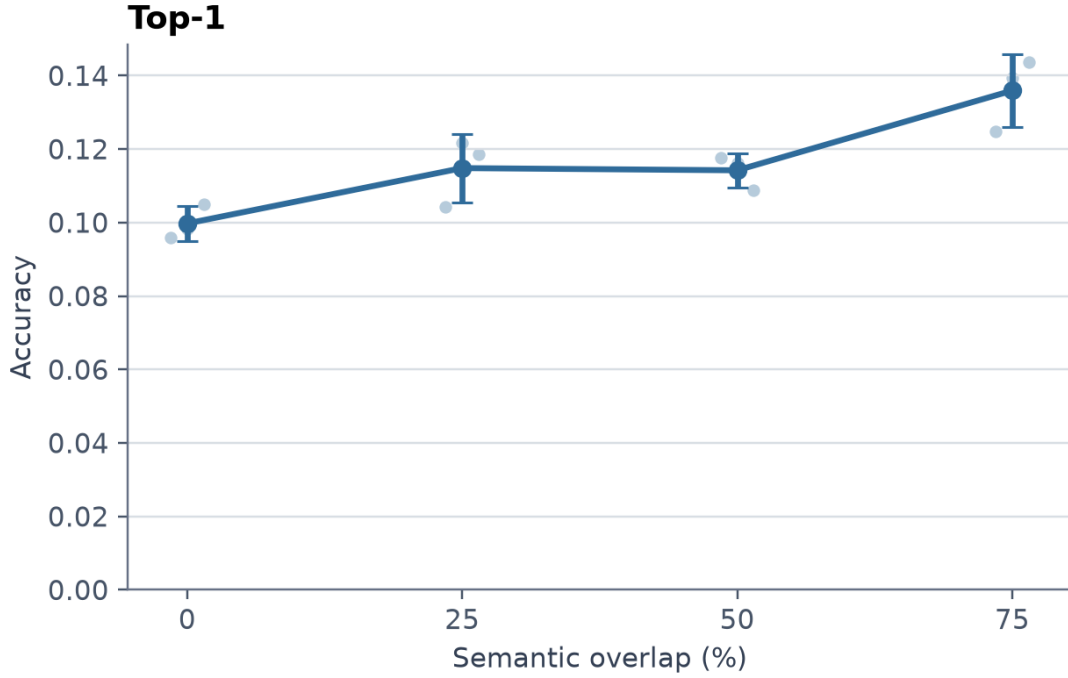


Figure 19: LM head result for semantic overlap.

G Significance Test

We use two tests, depending on the shape of the claim.

Comparison test. When an experiment asks whether one condition is more aligned than another, the metric is binary per item and the two conditions are evaluated on different items, so the comparison is unpaired. We use a one-sided Fisher’s exact test on the hits and misses of the two conditions ($\text{hits}_A, n_A, \text{hits}_B, n_B$), with the direction fixed in advance; a two-proportion z -test is reported for reference. For linear probing we take the counts at a single fixed layer (layer 3, a mid plateau layer, not the per-condition best). This test is used for the anchor necessity, punctuation, and special-token experiments.

Trend test. When an experiment varies an ordered quantity, we test the monotone trend with Spearman’s rank correlation between that quantity and the metric score. The unit is the condition: scores from repeated seeds are averaged into one point per condition rather than treated as independent samples. We report Spearman’s ρ as the effect size and a permutation p -value; with only four or five ordered points the p -value has a floor (around 0.04 and 0.017), so ρ carries the strength of the trend. This test is used for the graph density, semantic overlap, and lexical overlap experiments.